

COS20019 - Cloud Computing Architecture

## **Week 5: ACF Lab 5: RDS lab**

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Due Date: 08/10/2023

## Task 1: Create a Security Group for the RDS DB Instance

In this task, you will create a security group to allow your web server to access your RDS DB instance. The security group will be used when you launch the database instance.

5. In the **AWS Management Console**, on the Services menu, choose **VPC**.
6. In the left navigation pane, choose **Security Groups**.
7. Choose Create security group and then configure:
  - **Security group name:** DB Security Group
  - **Description:** Permit access from Web Security Group
  - **VPC:** Lab VPC
8. You will now add a rule to the security group to permit inbound database requests.
9. In the **Inbound rules** pane, choose Add rule  
The security group currently has no rules. You will add a rule to permit access from the *Web Security Group*.
10. Configure the following settings:
  - **Type:** MySQL/Aurora (3306)
  - **CIDR, IP, Security Group or Prefix List:** Type **sg** and then select *Web Security Group*.
11. This configures the Database security group to permit inbound traffic on port 3306 from any EC2 instance that is associated with the *Web Security Group*.
12. Choose Create security group  
You will use this security group when launching the Amazon RDS database.

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The screenshot shows the 'Create security group' page in the AWS VPC console. The browser address bar indicates the URL: `https://us-east-1.console.aws.amazon.com/vpc/home?region=us-east-1#CreateSecurityGroup`. The page title is 'Create security group'. Below the title, a brief description states: 'A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.'

The 'Basic details' section contains three input fields:

- Security group name:** 'DB Security Group'. A note below states: 'Name cannot be edited after creation.'
- Description:** 'Permit access from Web Security Group'.
- VPC:** 'vpc-0bbb0db7b91537156'.

The 'Inbound rules' section features a table with columns: 'Type', 'Protocol', 'Port range', 'Source', and 'Description - optional'. The 'Type' dropdown is set to 'MYSQL/Aurora'. The 'Protocol' dropdown is set to 'TCP'. The 'Port range' is '3306'. The 'Source' dropdown is set to 'Custom'. A search box next to it shows a selected rule ID: 'sg-0999b5186dbd251b3'. A 'Delete' button is visible next to the search box. An 'Add rule' button is at the bottom left of the table.

The screenshot shows the 'DB Security Group' details page in the AWS VPC console. The browser address bar indicates the URL: `https://us-east-1.console.aws.amazon.com/vpc/home?region=us-east-1#SecurityGroup:groupid=sg-03db5218db7e4f1d4`. A green banner at the top states: 'Security group (sg-03db5218db7e4f1d4 | DB Security Group) was created successfully'. Below the banner, the page title is 'sg-03db5218db7e4f1d4 - DB Security Group'. The 'Details' section contains a table with the following information:

| Details             |                      |                                       |                       |
|---------------------|----------------------|---------------------------------------|-----------------------|
| Security group name | Security group ID    | Description                           | VPC ID                |
| DB Security Group   | sg-03db5218db7e4f1d4 | Permit access from Web Security Group | vpc-0bbb0db7b91537156 |
| Owner               | Inbound rules count  | Outbound rules count                  |                       |
| 975463441896        | 1 Permission entry   | 1 Permission entry                    |                       |

## Task 2: Create a DB Subnet Group

In this task, you will create a *DB subnet group* that is used to tell RDS which subnets can be used for the database. Each DB subnet group requires subnets in at least two Availability Zones.

11. On the Services menu, choose **RDS**.
12. In the left navigation pane, choose **Subnet groups**.  
If the navigation pane is not visible, choose the menu icon in the top-left corner.
13. Choose Create DB Subnet Group then configure:
  - **Name:** DB-Subnet-Group
  - **Description:** DB Subnet Group
  - **VPC:** Lab VPC
14. Scroll down to the **Add Subnets** section.
15. Expand the list of values under **Availability Zones** and select the first two zones: **us-east-1a** and **us-east-1b**.
16. Expand the list of values under **Subnets** and select the subnets associated with the CIDR ranges **10.0.1.0/24** and **10.0.3.0/24**.  
These subnets should now be shown in the **Subnets selected** table.
17. Choose Create  
You will use this DB subnet group when creating the database in the next task.

Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations New

Events

Event subscriptions

Recommendations 0

Certificate update

RDS > Subnet groups > Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name

You won't be able to modify the name after your subnet group has been created.

DB-Subnet-Group

Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description

DB Subnet Group

VPC

Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.

Lab VPC (vpc-0bbb0db7b91537156)

Add subnets

Availability Zones

Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a X us-east-1b X

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

subnet-0eaed5de29974b8fa (10.0.1.0/24) X

subnet-0513f270418b177e1 (10.0.3.0/24) X

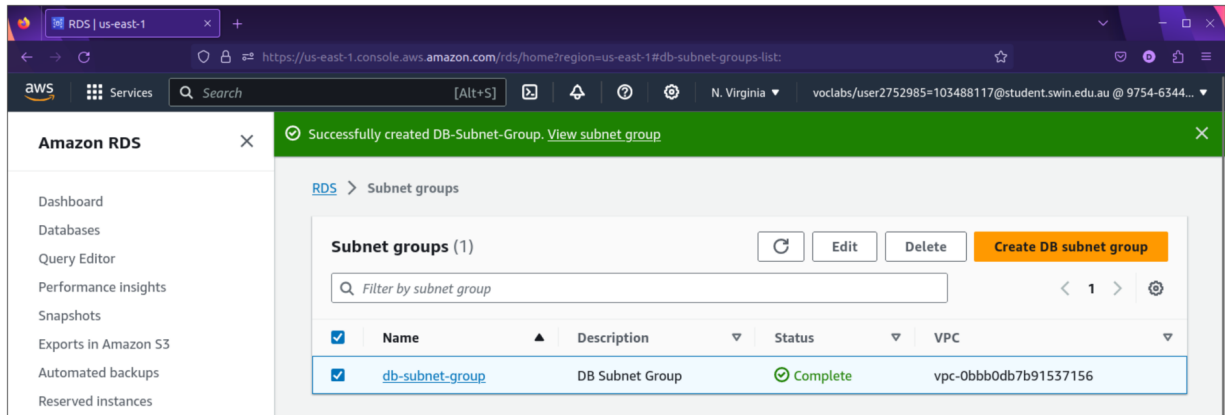
For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

| Availability zone | Subnet ID                | CIDR block  |
|-------------------|--------------------------|-------------|
| us-east-1a        | subnet-0eaed5de29974b8fa | 10.0.1.0/24 |
| us-east-1b        | subnet-0513f270418b177e1 | 10.0.3.0/24 |

CloudShell Feedback

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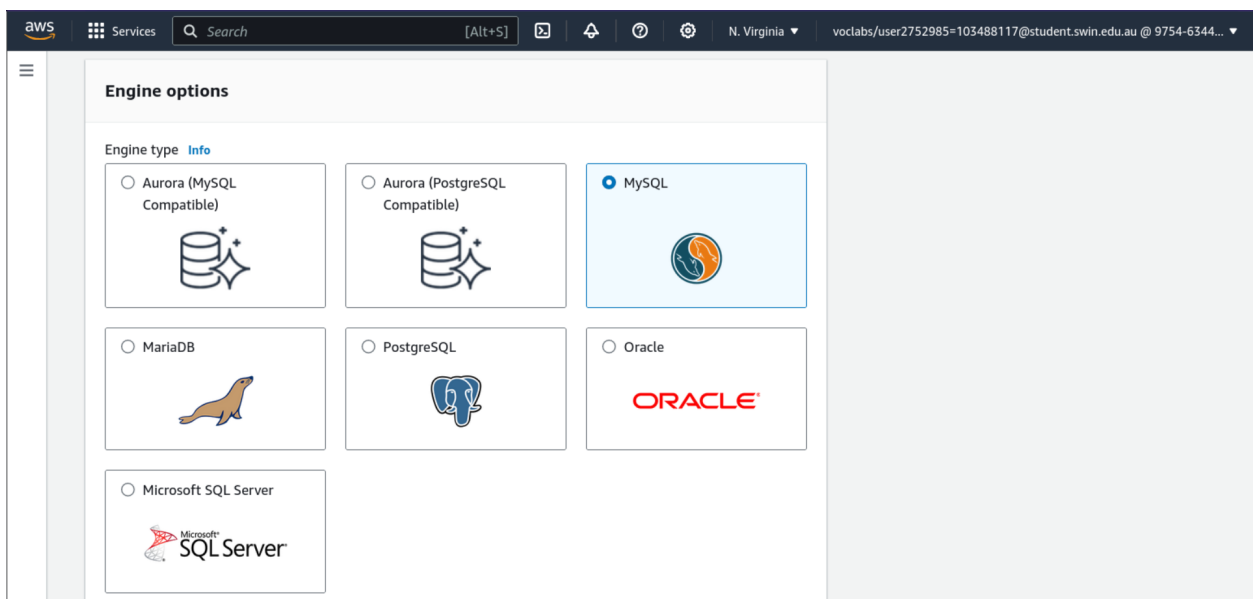


## Task 3: Create an Amazon RDS DB Instance

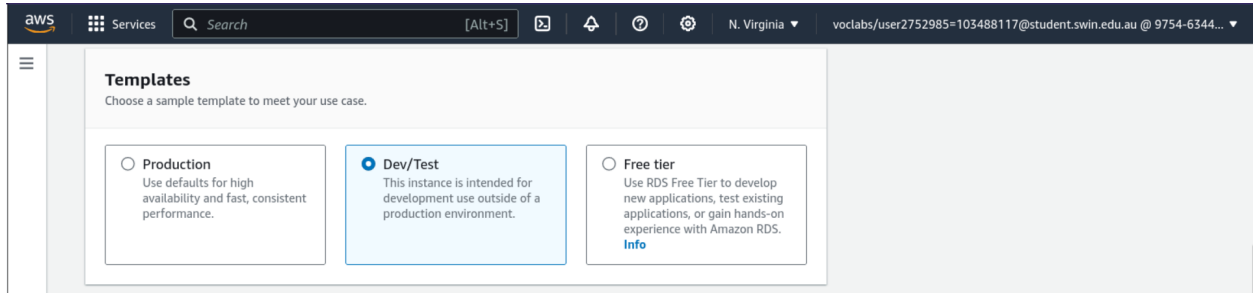
In this task, you will configure and launch a Multi-AZ Amazon RDS for MySQL database instance.

Amazon RDS **Multi-AZ** deployments provide enhanced availability and durability for Database (DB) instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

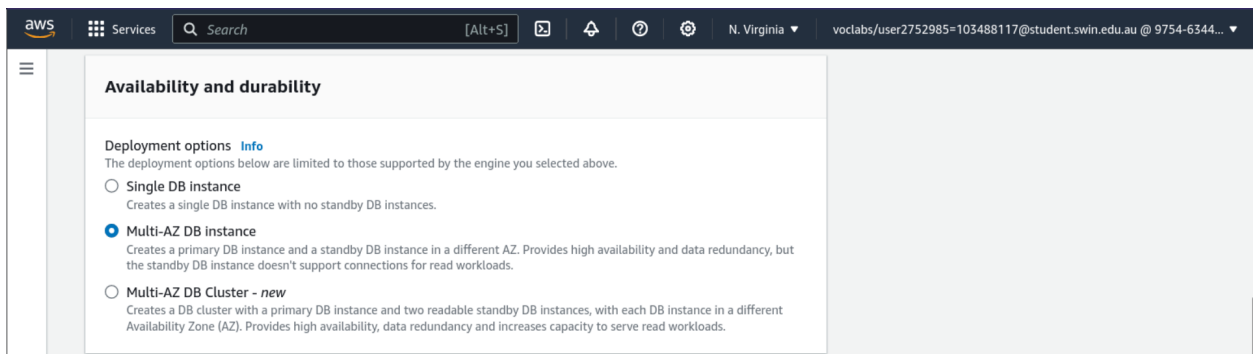
18. In the left navigation pane, choose **Databases**.
19. Choose Create database  
If you see **Switch to the new database creation flow** at the top of the screen, please choose it.
20. Select **MySQL** under **Engine Options**.



21. Under **Templates** choose **Dev/Test**.



22. Under **Availability and durability** choose **Multi-AZ DB instance**.



23. Under **Settings**, configure:

- **DB instance identifier:** lab-db
- **Master username:** main
- **Master password:** lab-password
- **Confirm password:** lab-password

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The screenshot shows the 'Settings' page in the AWS Management Console. The 'DB instance identifier' is set to 'lab-db'. Under 'Credentials Settings', the 'Master username' is 'main'. The checkbox 'Manage master credentials in AWS Secrets Manager' is unchecked. A blue information box states: 'If you manage the master user credentials in Secrets Manager, some RDS features aren't supported. Learn more'. The 'Auto generate a password' checkbox is also unchecked. The 'Master password' and 'Confirm master password' fields are empty.

**Settings**

**DB instance identifier** [Info](#)  
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.  
  
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ **Credentials Settings**

**Master username** [Info](#)  
Type a login ID for the master user of your DB instance.  
  
1 to 16 alphanumeric characters. The first character must be a letter.

☐ **Manage master credentials in AWS Secrets Manager**  
Manage master user credentials in Secrets Manager. RDS can generate a password for you and manage it throughout its lifecycle.

**Info** If you manage the master user credentials in Secrets Manager, some RDS features aren't supported. [Learn more](#)

☐ **Auto generate a password**  
Amazon RDS can generate a password for you, or you can specify your own password.

**Master password** [Info](#)  
  
Constraints: At least 8 printable ASCII characters. Can't contain any of the following: / (slash), ' (single quote), " (double quote) and @ (at sign).

**Confirm master password** [Info](#)

24. Under **DB instance class**, configure:

- Select **Burstable classes (includes t classes)**.
- Select *db.t3.micro*

The screenshot shows the 'Instance configuration' page. A blue information box at the top says: 'Amazon RDS Optimized Writes - new Info' with a toggle switch for 'Show instance classes that support Amazon RDS Optimized Writes'. Under 'DB instance class', the radio button 'Burstable classes (includes t classes)' is selected. The dropdown menu shows 'db.t3.micro' with details: '2 vCPUs 1 GiB RAM Network: 2,085 Mbps'. The 'Include previous generation classes' checkbox is unchecked.

**Instance configuration**  
The DB instance configuration options below are limited to those supported by the engine that you selected above.

**Amazon RDS Optimized Writes - new** [Info](#)  
☐ Show instance classes that support Amazon RDS Optimized Writes

**DB instance class** [Info](#)

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ **Burstable classes (includes t classes)**

2 vCPUs 1 GiB RAM Network: 2,085 Mbps ▼

☐ Include previous generation classes

25. Under **Storage**, configure:

- **Storage type:** *General Purpose (SSD)*
- **Allocated storage:** *20*



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The screenshot shows the AWS Management Console 'Storage' configuration page. The 'Storage type' is set to 'General Purpose SSD (gp3)' with a note that performance scales independently from storage. The 'Allocated storage' is set to '20' GiB, with a range from 20 to 6,144 GiB. A blue information box states: 'After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. Learn more'. Below this, 'Advanced settings' are expanded, showing 'Baseline IOPS of 3,000 IOPS and storage throughput of 125 MiBps are included for allocated storage less than 400 GiB.' and 'Storage autoscaling' is also listed.

26. Under **Connectivity**, configure:

- **Virtual Private Cloud (VPC):** *Lab VPC*

The screenshot shows the AWS Management Console 'Connectivity' configuration page. Under 'Compute resource', the 'Don't connect to an EC2 compute resource' option is selected. Under 'Virtual private cloud (VPC)', the 'Lab VPC (vpc-0bbb0db7b91537156)' is selected from a dropdown menu. A note below the dropdown states: 'Only VPCs with a corresponding DB subnet group are listed.'

27. Under **Existing VPC security groups**, from the dropdown list:

- Choose *DB Security Group*.
- Deselect *default*.

The screenshot shows the AWS Management Console 'VPC security group (firewall)' configuration page. The 'Choose existing' option is selected. In the 'Existing VPC security groups' dropdown, 'DB Security Group' is selected and shown in a tag with a close button (X).

28. Expand **Additional configuration**, then configure:

- **Initial database name:** **lab**
- Uncheck **Enable automatic backups**.
- Uncheck **Enable encryption**
- Uncheck **Enable Enhanced monitoring**.

The screenshot shows the 'Additional configuration' section of the AWS Management Console. It includes the following settings:

- Database options:**
  - Initial database name:
  - DB parameter group:
  - Option group:
- Backup:**
  - ☐ **Enable automated backups**  
Creates a point-in-time snapshot of your database.
- Encryption:**
  - ☐ **Enable encryption**  
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console.

The screenshot shows the 'Monitoring' section of the AWS Management Console. It includes the following setting:

- Monitoring:**
  - ☐ **Enable Enhanced monitoring**  
Enabling Enhanced monitoring metrics are useful when you want to see how different processes or threads use the CPU.

29. This will turn off backups, which is not normally recommended, but will make the database deploy faster for this lab.
30. Choose Create database  
Your database will now be launched.  
If you receive an error that mentions "not authorized to perform: iam:CreateRole", make sure you unchecked *Enable Enhanced monitoring* in the previous step.
31. Choose **lab-db** (choose the link itself).  
You will now need to wait **approximately 4 minutes** for the database to be available. The deployment process is deploying a database in two different Availability zones.  
While you are waiting, you might want to review the [Amazon RDS FAQs](#) or grab a cup of coffee.
32. Wait until **Info** changes to **Modifying** or **Available**.

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The screenshot shows the Amazon RDS console. A green notification banner at the top states "Successfully created database lab-db" and provides instructions on using settings to simplify configuration. Below the notification, the "Databases" section is visible, showing a table with one database instance: "lab-db". The instance is in the "Available" state, using the "MySQL Community" engine, located in the "us-east-1b" region, with a size of "db.t3.micro". The table has columns for "DB identifier", "Status", "Role", "Engine", "Region & AZ", "Size", and "Actions".

33. Scroll down to the **Connectivity & security** section and copy the **Endpoint** field.  
It will look similar to: *lab-db.cggq8lhnxvnx.us-west-2.rds.amazonaws.com*

The screenshot shows the Amazon RDS console for the "lab-db" instance. The "Connectivity & security" section is expanded, displaying the following details:

| Summary       |                  |                 |             |
|---------------|------------------|-----------------|-------------|
| DB identifier | CPU              | Status          | Class       |
| lab-db        | 4.02%            | Available       | db.t3.micro |
| Role          | Current activity | Engine          | Region & AZ |
| Instance      | 0 Connections    | MySQL Community | us-east-1b  |

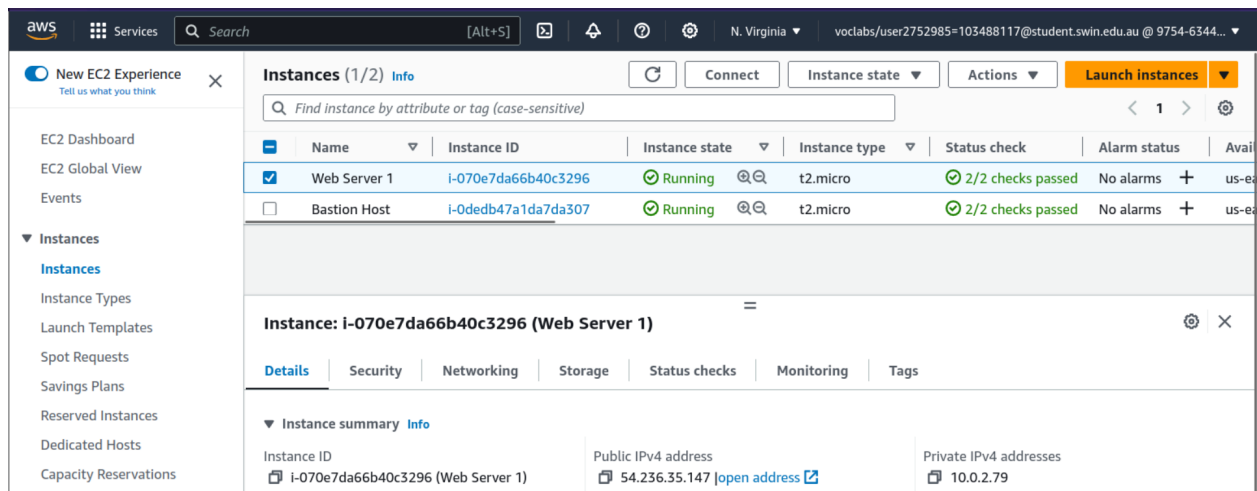
| Connectivity & security                                                                                                           |                                                                                                |                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Endpoint &amp; port</b>                                                                                                        | <b>Networking</b>                                                                              | <b>Security</b>                                                                               |
| Endpoint<br><a href="https://lab-db.cggq8lhnxvnx.us-east-1.rds.amazonaws.com">lab-db.cggq8lhnxvnx.us-east-1.rds.amazonaws.com</a> | Availability Zone<br>us-east-1b                                                                | VPC security groups<br><a href="#">DB Security Group (sg-03db5218db7e4f1d4)</a><br>Active     |
| Port<br>3306                                                                                                                      | VPC<br><a href="#">Lab VPC (vpc-0bbb0db7b91537156)</a>                                         | Publicly accessible<br>No                                                                     |
|                                                                                                                                   | Subnet group<br>db-subnet-group                                                                | Certificate authority<br><a href="#">Info</a><br>rds-ca-2019                                  |
|                                                                                                                                   | Subnets<br><a href="#">subnet-0513f270418b177e1</a><br><a href="#">subnet-0eae5de29974b8fa</a> | Certificate authority date<br>August 23, 2024, 00:08 (UTC+07:00)                              |
|                                                                                                                                   | Network type<br>IPv4                                                                           | DB instance certificate expiration date<br><a href="#">August 23, 2024, 00:08 (UTC+07:00)</a> |

34. Paste the Endpoint value into a text editor. You will use it later in the lab.

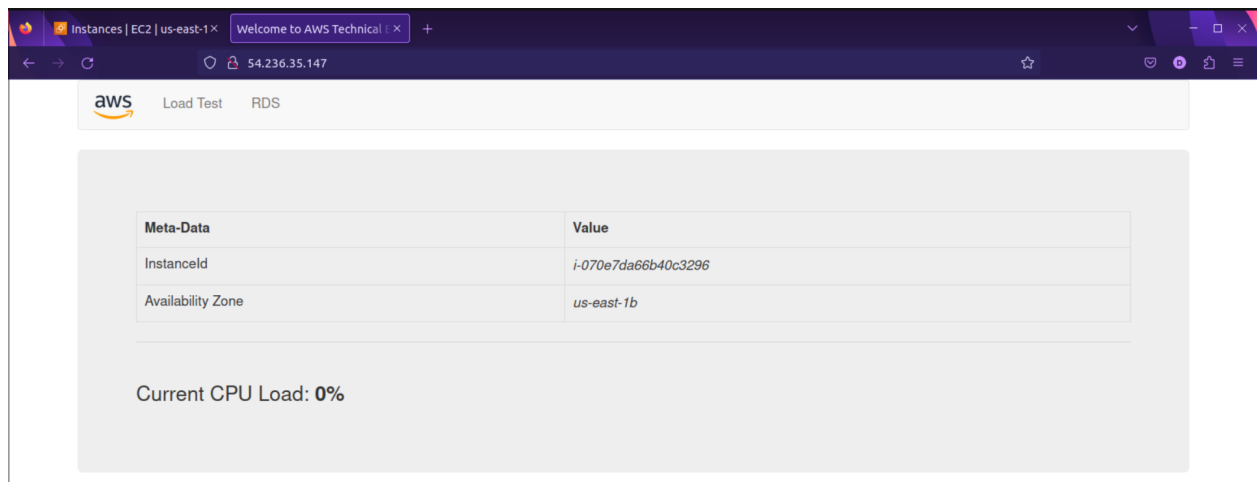
## Task 4: Interact with Your Database

In this task, you will open a web application running on your web server and configure it to use the database.

32. To copy the **WebServer** IP address, choose on the Details drop down menu above these instructions, and then choose Show.



33. Open a new web browser tab, paste the *WebServer* IP address and press Enter.  
The web application will be displayed, showing information about the EC2 instance.



34. Choose the **RDS** link at the top of the page.  
You will now configure the application to connect to your database.
35. Configure the following settings:
- **Endpoint:** Paste the Endpoint you copied to a text editor earlier
  - **Database:** lab
  - **Username:** main

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- Password: lab-password
- Choose Submit

 Load Test RDS

Endpoint

lab-db.cqehgqx7m1k.us-east-1.rds.amazonaws.com

Database

lab

Username

main

Password

\*\*\*\*\*

Submit

A message will appear explaining that the application is running a command to copy information to the database. After a few seconds the application will display an **Address Book**. The Address Book application is using the RDS database to store information.

 Load Test RDS

### Address Book

| Last name | First name | Phone        | Email                      | Admin                |                        |
|-----------|------------|--------------|----------------------------|----------------------|------------------------|
| Doe       | Jane       | 010-110-1101 | janed@someotheraddress.org | <a href="#">Edit</a> | <a href="#">Remove</a> |
| Johnson   | Roberto    | 123-456-7890 | robertoj@someaddress.com   | <a href="#">Edit</a> | <a href="#">Remove</a> |

36. Test the web application by adding, editing and removing contacts.

The data is being persisted to the database and is automatically replicating to the second Availability Zone.

 Load Test RDS

### Address Book

| Last name | First name   | Phone          | Email                      | Admin                       |                        |
|-----------|--------------|----------------|----------------------------|-----------------------------|------------------------|
|           |              |                |                            | <a href="#">Add Contact</a> |                        |
| Doe       | Jane         | 010-110-1101   | janed@someotheraddress.org | <a href="#">Edit</a>        | <a href="#">Remove</a> |
| Johnson   | Roberto      | 123-456-7890   | robertoj@someaddress.com   | <a href="#">Edit</a>        | <a href="#">Remove</a> |
| Luong     | Trac Duc Anh | 84-393-563-123 | luongtrac.ducanh@gmail.com | <a href="#">Edit</a>        | <a href="#">Remove</a> |